IAT New Test 4

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

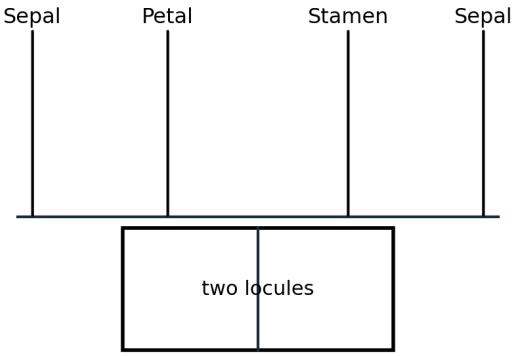
Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 A woody plant bears pollen and ovules on separate cones; its ovules are exposed at pollination, although a seed coat develops later. Which observation distinguishes it from an angiosperm?
A) Ovules are not enclosed in an ovary before fertilisation
B) It produces pollen grains
C) Its diploid zygote develops into an embryo
D) It forms vascular tissue

Q2 The figure depicts a floral section: sepals, petals and stamens arise above the ovary; the ovary has two chambers. If the flower is turned upside down for display, which classification remains valid?



Ovary below whorl attachment

A) Superior, unilocular ovary
B) Inferior, bilocular ovary
C) Superior, bilocular ovary
D) Inferior, unilocular ovary

Q3 Complete hydrolysis of one unbranched peptide releases 18 amino-acid molecules. How many water molecules were consumed if no cyclic bond was present?
A) 18
B) 16
C) 17
D) 19

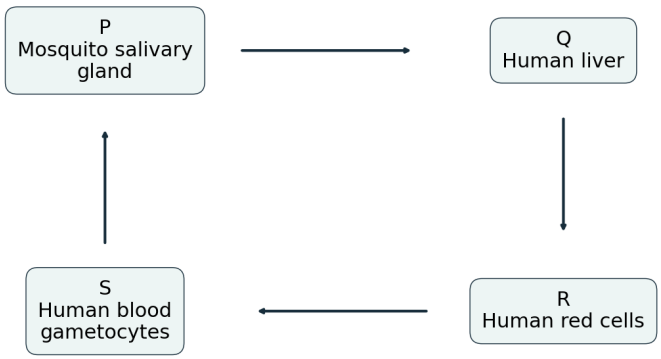
Q4 A tissue consumes 20 mL O_2 and releases 16 mL CO_2 in a short closed-chamber trial. What is its measured respiratory quotient, and which pure-substrate value is it closer to: glucose 1.0 or a typical fat 0.7?
A) 1.25, glucose
B) 1.25, fat
C) 0.8, glucose
D) 0.8, fat

Q5 After a booster dose of the same antigen, serum antibody rises sooner and higher than after the first exposure. Which cellular population most directly explains the difference?
A) Long-lived memory B cells selected during the first exposure
B) Mature erythrocytes that bind antigen
C) Platelets that secrete antigen-specific IgG
D) Neutrophils converting into plasma cells

Q6 In an angiosperm, a fertilised ovule develops normally but the surrounding ovary wall is experimentally removed immediately after fertilisation. Which structure can still arise directly from the ovule?
A) Pericarp
B) Seed with embryo and seed coat
C) Fruit wall
D) Receptacle

Q7 An allele begins at frequency 0.10 in eight large replicate populations. After many generations under the same insecticide, it rises above 0.80 in all eight; without insecticide, frequencies fluctuate around 0.10. What is the strongest inference?
A) Genetic drift must cause the consistent directional rise
B) Insecticide directly edits the allele in every individual
C) The data support selection favouring the allele under insecticide
D) The allele is necessarily dominant

Q8 The figure shows a simplified malaria cycle with stages P in mosquito salivary glands, Q in human liver cells, R in human red cells and S as gametocytes in human blood. Which stage enters a human with the mosquito bite?



A) P, sporozoites
B) Q, liver schizonts
C) R, merozoites in erythrocytes
D) S, gametocytes

Q9 A plasmid carries antibiotic-resistance genes ampR and tetR; foreign DNA interrupts tetR. Colonies grow on ampicillin but fail to grow on tetracycline. What does this pattern support?
A) No plasmid entered the cells
B) A recombinant plasmid disrupted tetR
C) Both resistance genes remain intact
D) The insert disrupted ampR

Q10 Producers capture 80,000 kJ; approximately 10% of available energy transfers at each trophic step. What energy reaches tertiary consumers through producers→herbivores→secondary→tertiary consumers?
A) 8,000 kJ
B) 800 kJ
C) 80 kJ
D) 8 kJ

Q11 A dihybrid testcross yields four phenotype classes of 98, 102, 101 and 99 offspring. Which conclusion best fits these counts?
A) Complete linkage of the two loci
B) A 9:3:3:1 ratio from two heterozygotes
C) One locus is lethal in homozygotes
D) Approximate independent assortment of the two loci

Q12 Two unrelated unaffected people each have an affected sibling with the same fully penetrant autosomal-recessive disorder. Each person's parents are known Aa × Aa. The couple's first child is unaffected. With independent family origins and no new mutation, what is the probability their second child is affected?
A) 3/32
B) 1/9
C) 1/16
D) 1/4

Q13 Heart rate rises from 60 to 90 beats/min while stroke volume falls from 80 to 60 mL/beat. What is the change in cardiac output?
A) Decrease by 0.6 L/min
B) Increase by 0.6 L/min
C) Increase by 2.4 L/min
D) No change

Q14 An aquaporin increases water permeability of a membrane without consuming ATP. Which observation is compatible with this channel?
A) Water always moves from higher solute concentration to lower
B) Aquaporin pumps solutes against their gradient
C) Net water movement follows a water-potential gradient
D) Transport stops whenever ATP synthesis stops

Q15 For one region S=cA^z. Species counts are 20 and 40 at areas A and 4A, respectively. What is z?
A) 1
B) 2
C) 1/4
D) 1/2

Chemistry

Q16 In the displayed structure $CH_2{=}CH{-}C{\equiv}N$, how many σ and π bonds are present, including C–H bonds?



A) 6 σ and 3 π
B) 5 σ and 3 π
C) 6 σ and 2 π
D) 7 σ and 2 π

Q17 A complex ion $[M(NH_3)_4Cl_2]^+$ contains neutral NH_3 and chloride ligands. What is M's oxidation state and its coordination number?
A) +1, 6
B) +3, 6
C) +2, 4
D) +3, 4

Q18 In one valid Lewis resonance contributor of NO_3^- with one N=O and two N–O single bonds, what are the formal charges on N and each singly bonded O?
A) N 0; each O −1
B) N +1; each O 0
C) N −1; each O +1
D) N +1; each O −1

Q19 A hydrogen atom in an excited level emits exactly two photons in succession and ends at n=1. Their wavenumbers are R(1/4−1/16) and R(1−1/4), in that order. Which sequence of levels produced them?
A) 4→2→1
B) 3→2→1
C) 4→3→1
D) 5→2→1

Q20 A secondary carbocation can undergo a 1,2-hydride shift to a tertiary carbocation without changing total molecular formula. Which statement best predicts the change?
A) It necessarily forms a less stable primary ion
B) The rearrangement can be favourable because hyperconjugation stabilises the tertiary centre
C) Hydride transfer adds a hydrogen atom from solvent
D) A hydride shift changes the net positive charge to negative

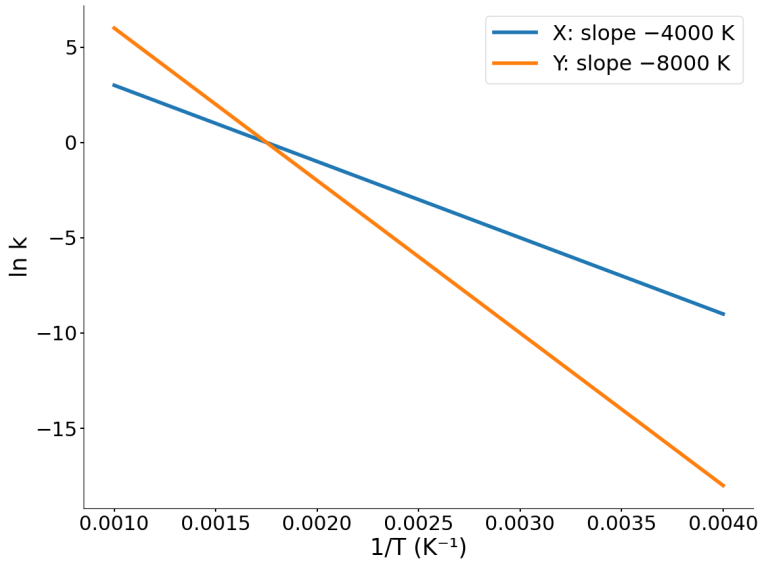
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Q21 Which product pair forms when CH_3COCl undergoes complete hydrolysis in water?
A) CH_3CHO and Cl_2
B) CH_3OH and HCl
C) CH_3COOH and HCl
D) CH_3COCH_3 and H_2O

Q22 An amine reacts with benzenesulfonyl chloride; its sulfonamide product dissolves in aqueous NaOH but reprecipitates upon acidification. Which class of starting amine is indicated?
A) Primary amine
B) Secondary amine
C) Tertiary amine
D) Quaternary ammonium salt

Q23 A neutral-side-chain amino acid has $pK_a(COOH)=2$ and $pK_a(NH_3^+)=9$. At pH 6, which form predominates?
A) H_3N^+–CHR–COOH, net +1
B) H_3N^+–CHR–COO^-, net 0
C) H_2N–CHR–COO^-, net −1
D) H_2N–CHR–COOH, net 0

Q24 The figure plots ln k against 1/T for reactions X and Y with straight lines of slopes −4000 K and −8000 K respectively. Assuming the same units and Arrhenius behaviour, which has the greater activation energy?



A) X, by a factor of two
B) X and Y are equal
C) Slopes alone cannot compare activation energies
D) Y, by a factor of two

Q25 Equal volumes of 1.0×10^{-4} M $AgNO_3$ and 1.0×10^{-6} M NaCl are mixed. For AgCl, $K_{sp} = 1.0\times10^{-10}$. Neglect other reactions and activity effects. Which conclusion follows immediately after mixing, before any precipitation?
A) Precipitation, because $Q = 10^{-10}$
B) Precipitation, because $Q = 10^{-8}$
C) No precipitation, because $Q = 2.5 \times 10^{-11}$
D) No precipitation, because $Q = 2.5 \times 10^{-9}$

Q26 Chlorine reacts with cold dilute NaOH to produce Cl^- and ClO^-. Which electron-accounting statement is consistent with the balanced reaction?
A) Both chlorine atoms are reduced from 0 to −1
B) One chlorine atom loses one electron while another gains one electron
C) Chlorine's average product oxidation state is +1
D) Oxygen is oxidised from −2 to 0

Q27 Equal masses of the same liquid at 20 °C and 80 °C are mixed in an insulated vessel with negligible heat capacity and no phase change. What is the final temperature?
A) 20 °C
B) 80 °C
C) 60 °C
D) 50 °C

Q28 An ideal liquid mixture contains mole fraction xA=0.25. Pure vapour pressures at this temperature are pA°=80 kPa and pB°=40 kPa. What is A's fraction of the total vapour pressure?
A) 1/2
B) 1/3
C) 2/3
D) 2/5

Q29 Nitration of toluene gives mostly ortho and para products. What change to the ring substituent would instead favour meta nitration?
A) Replace CH_3 with NO_2
B) Replace CH_3 with OCH_3
C) Replace CH_3 with NH_2
D) Replace CH_3 with Cl

Q30 At room temperature, benzyl chloride undergoes nucleophilic substitution readily, while chlorobenzene resists ordinary SN1/SN2 substitution. Which distinction best explains this?
A) Benzyl chloride lacks a C–Cl bond
B) Chlorobenzene forms a more stable phenyl cation than a benzyl cation
C) Chlorobenzene's aryl C–Cl bond has partial double-bond character and its carbon is sp^2
D) A benzyl carbon cannot undergo backside attack

Mathematics

Q31 A hyperbola centred at the origin has transverse vertices $(\pm 3,0)$ and foci $(\pm 5,0)$. Which pair of lines are its asymptotes?
A) $y=\pm(4/3)x$
B) $y=\pm(3/4)x$
C) $y=\pm(5/3)x$
D) $y=\pm(4/5)x$

Q32 Every line $(k+1)x+(2k-1)y=3k+2$ passes through one fixed point. What is that point?
A) $(1,1)$
B) $(7/3,1/3)$
C) $(2,0)$
D) $(1,2)$

Q33 Evaluate $\int$ from 0 to 1 of $(2x+3)/[(x+1)(x+2)]\,dx$.
A) $\ln 2$
B) $\ln(3/2)$
C) $\ln 3$
D) $\ln 6$

Q34 What is the area enclosed by $y=x^2$ and $y=2x$ between their intersections?
A) 1/3
B) 2/3
C) 8/3
D) 4/3

Q35 For $x^2+xy+y^2=7$, the point $(1,2)$ lies on the curve. What is the slope of its tangent there?
A) $-4/5$
B) $-5/4$
C) 4/5
D) 0

Q36 On integers define aRb when 4 divides $(a-b)$. If the domain is restricted to odd integers, how many equivalence classes remain?
A) 1
B) 4
C) 2
D) Infinitely many

Q37 Let $A=[[1,1],[0,1]]$, $B=[[1,0],[1,1]]$, and $AX=B$. What is $\mathrm{tr}(X)$?
A) 3
B) 2
C) 0
D) 1

Q38 A fair coin is tossed six times. Given that exactly three heads occurred, what is the probability that there is exactly one adjacent HH pair? Overlapping pairs count separately, so HHH contains two HH pairs.
A) 1/5
B) 3/5
C) 2/5
D) 4/5

Q39 For acute α and β, tan α=1/2 and tan β=1/3. What is cos(α+β)?
A) 1/√10
B) 3/√50
C) 1/√2
D) 2/√5

Q40 What is the foot of the perpendicular from P(2,−3,5) to the xy-plane?
A) (0,−3,5)
B) (2,0,5)
C) (0,0,5)
D) (2,−3,0)

Q41 Vectors a=(1,2,0) and b=(2,1,0) form two sides of a parallelogram. Which vector gives its oriented area a×b?
A) (0,0,−3)
B) (0,0,3)
C) (3,3,0)
D) (0,0,5)

Q42 Let ω be a non-real cube root of unity. What is $|1-\omega|^2$?
A) 1
B) 2
C) 3
D) 4

Q43 For which real t does $x^2-tx+4=0$ have two distinct positive real roots?
A) t>0
B) t≥4
C) t<−4
D) t>4

Q44 For positive integer p in {1,2,4,10}, which p makes the constant-term coefficient of $(x^2+x\text{^}(-p))^6$ largest?
A) 2
B) 1
C) 4
D) 10

Q45 The infinite geometric series $1+r+r^2+\cdots$ has a finite sum for which condition on real r?
A) r≤1
B) |r|<1
C) |r|≤1
D) r>−1

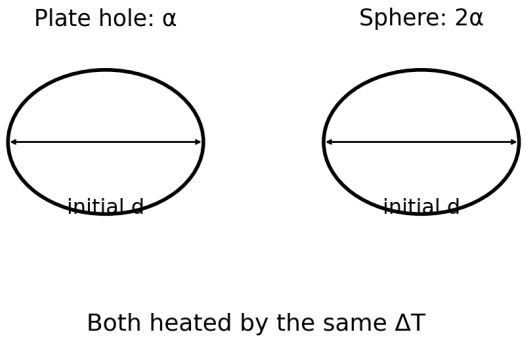
Physics

Q46 Voltage and current are measured independently with maximum percentage uncertainties 2% and 1%. For R=V/I, what is the maximum first-order percentage uncertainty in R?
A) 3%
B) 2%
C) 1%
D) 0.02%

Q47 A 5 kg block rests on a horizontal rough floor with μs=0.4. Taking g=10 m/s², what horizontal force first reaches the limiting static friction?
A) 10 N
B) 20 N
C) 50 N
D) 2 N

Q48 Two wires of equal length L and cross-section A, with Young moduli Y and 2Y, are joined end to end and pulled by force F. What is their combined extension?
A) FL/(2AY)
B) FL/(AY)
C) 2FL/(AY)
D) 3FL/(2AY)

Q49 In the figure, a metal sphere just fits through a circular hole in a metal plate at 20 °C. Both are heated equally; the sphere's expansion coefficient is twice the plate's. What happens for a small temperature rise?



A) The hole contracts while the sphere grows
B) Both sizes grow at exactly the same fractional rate
C) The sphere can no longer pass through the hole
D) The hole grows twice as fast fractionally

Q50 A rigid insulated container holds one mole of ideal diatomic gas with vibrational modes inactive. A paddle does $5RT_0/2$ work on gas initially at T_0, with no heat transfer. What is final temperature?
A) $3T_0/2$
B) $5T_0/2$
C) $3T_0$
D) $2T_0$

Q51 A small-amplitude pendulum is lengthened from L to 1.44L while g stays fixed. What is the ratio of new to old period?
A) 1.44
B) 1.20
C) 0.83
D) 2.07

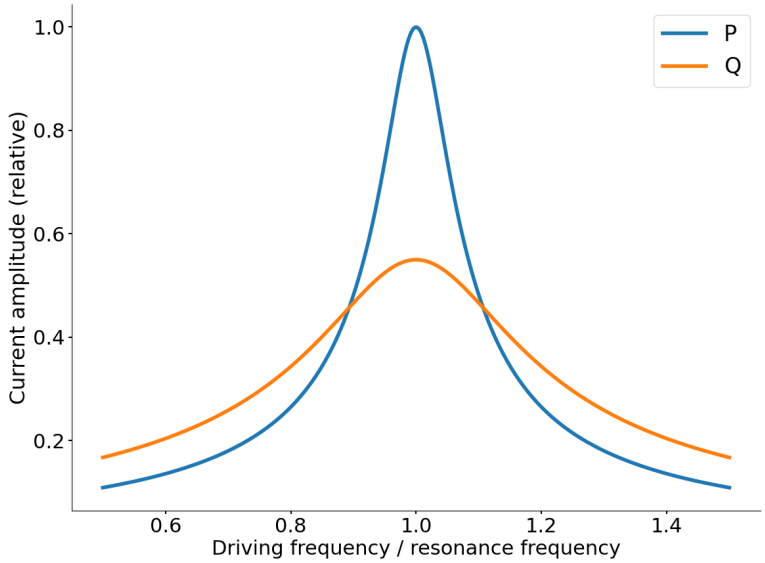
Q52 An unknown tone gives 4 beats/s with 256 Hz and 8 beats/s with 260 Hz. What is its frequency?
A) 252 Hz
B) 264 Hz
C) 260 Hz
D) 256 Hz

Q53 In $0<x<2$ m a uniform electric field is +3 V/m along x. If $V(0)=10$ V, what is $V(2)$?
A) 16 V
B) 10 V
C) 7 V
D) 4 V

Q54 A wire with carrier density n and drift speed v has current density J. A second wire has carrier density n/2 and drift speed 2v, with the same charge magnitude. What is its current density?
A) J/4
B) J
C) 2J
D) 4J

Q55 An ideal 2.0 H inductor and 50 µF capacitor form a closed lossless circuit. Initially the capacitor is at 100 V and current is zero. At a later instant the current magnitude is 0.40 A. What is the capacitor voltage magnitude then?
A) 40 V
B) 80 V
C) 60 V
D) 100 V

Q56 The figure compares current amplitude versus driving frequency for series RLC circuits P and Q with the same L and C. P has a taller, narrower resonance peak than Q. Which is consistent?



A) P has greater resistance and greater bandwidth
B) P has greater resistance but smaller bandwidth
C) Both necessarily have different resonance frequencies
D) P has smaller resistance and smaller bandwidth

Q57 A prism has angle 60° and minimum deviation 30° in air. What is its refractive index?
A) $\sqrt{2}$
B) $\sqrt{3}$
C) 2
D) $1/\sqrt{2}$

Q58 A film of refractive index n lies between air and a glass substrate of index greater than n. At normal incidence, what condition gives constructive interference of the two principal reflected rays, neglecting additional reflections?
A) $2nt=(m+1/2)\lambda_0$ because one reflection reverses phase
B) $2nt=m\lambda_0$ because both reflections reverse phase
C) $t=m\lambda_0$ regardless of n
D) Every thickness gives destructive reflection

Q59 Nonrelativistic electrons accelerated from rest through V have wavelength λ. What wavelength results from acceleration through 4V?
A) 4λ
B) 2λ
C) $\lambda/2$
D) $\lambda/4$

Q60 An ideal half-wave rectifier sends $v(t)=V_0 \sin\omega t$ to a resistor R during the positive half-cycle and zero otherwise. What is average resistor power over a full cycle?
A) V_0^2/R
B) $V_0^2/(2R)$
C) $V_0^2/(4R)$
D) $V_0^2/(8R)$

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / A Exposed ovules in gymnosperms

Answer: (A) Ovules are not enclosed in an ovary before fertilisation.

Why: Naked ovules on cone scales identify the gymnosperm condition. A subsequent seed coat does not turn the cone into a fruit, which requires an ovary wall. Pollen and embryo formation alone do not separate gymnosperms from angiosperms. The diagnostic feature is the exposed ovule at pollination; a seed coat forms later without an enclosing fruit wall.

Why not the others: (B) Pollen, (C) embryos and (D) vascular tissue occur in both seed-plant groups, so each fails the comparison.

Key point: Compare the ovule's enclosure at pollination, not merely the presence of seeds.

Q2 / B Floral whorls and ovary position

Answer: (B) Inferior, bilocular ovary.

Why: The insertion level of other whorls is above the ovary, defining an epigynous flower with an inferior ovary; a septum divides it into two locules. Rotating a drawing changes its page orientation, not these structural relationships. The diagram's attachment line lies above the drawn ovary, and the internal septum divides two spaces. Turning the paper upside down leaves both anatomical relationships unchanged; labels such as superior refer to whorl insertion, not page direction.

Why not the others: (A) and (C) put the attachment below the ovary; (A) and (D) ignore the septum.

Key point: Infer position from attachment, and locule count from internal partitions.

Q3 / C Polypeptide hydrolysis and peptide bonds

Answer: (C) 17.

Given: The unbranched chain contains 18 residues.

Formula/Principle: A linear chain of n residues contains n−1 peptide linkages; hydrolysis consumes one water per linkage.

Step 1: Count 18−1=17 bonds.

Step 2: Breaking each bond uses one H_2O, giving 17 molecules.

Why: The terminal residues do not require a bond beyond the chain ends. A linear chain with n amino-acid residues has $n-1$ peptide links because each link joins a pair of neighbours. Hydrolysing each link adds one water, so the ends do not contribute an eighteenth cleavage.

Why not the others: (A) counts residues rather than links; (B) subtracts twice; (D) adds a nonexistent terminal bond.

Key point: Distinguish monomer number from linkage number.

Q4 / D Respiratory quotient and substrate mixture

Answer: (D) 0.8, fat.

Given: CO_2 evolved=16 mL and O_2 consumed=20 mL.

Formula/Principle: RQ=volume CO_2/volume O_2 under comparable conditions.

Step 1: RQ=16/20=0.8.

Step 2: Distance to 1.0 is 0.2, while distance to 0.7 is 0.1, so this reading is closer to the fat reference.

Why: The observed ratio may reflect a mixture of substrates; closeness alone does not prove the exclusive substrate. An RQ between the pure-glucose and typical-fat benchmarks is compatible with mixed metabolism or other substrate use. The 0.8 measurement is closer numerically to 0.7, but that proximity alone does not chemically identify a pure fat fuel.

Why not the others: (A) and (B) invert the ratio; (C) misjudges the closeness.

Key point: Compute the ratio before comparing reference values.

Q5 / A Primary versus secondary immune response

Answer: (A) Long-lived memory B cells selected during the first exposure.

Why: Priming selects and expands antigen-specific clones. Memory cells respond rapidly on re-exposure and differentiate into antibody-secreting cells, explaining both shorter lag and greater magnitude. The secondary response begins with more antigen-experienced B-cell clones and can also benefit from helper-cell memory. The higher titre therefore reflects altered immune-cell history rather than a change in erythrocytes or platelets.

Why not the others: (B) Erythrocytes do not mediate this clonal response; (C) platelets are not antibody producers; (D) neutrophils do not become B-lineage plasma cells.

Key point: The booster changes the available antigen-specific cell population.

Q6 / B Double fertilisation and seed tissues

Answer: (B) Seed with embryo and seed coat.

Why: The ovule gives rise to the seed, its integuments form the seed coat, and the zygote forms the embryo. Removal of the ovary wall prevents normal pericarp development, not the ovule's identity. Double fertilisation and subsequent ovule development can continue in the stipulated experiment. The embryo derives from the zygote, and the seed coat from integuments; only the removed ovary wall would normally supply pericarp.

Why not the others: (A) and (C) describe the ovary-derived fruit wall; (D) is a floral axis, not an ovule derivative.

Key point: Trace seed and fruit structures to different floral precursors.

Q7 / C Selection versus drift in replicate populations

Answer: (C) The data support selection favouring the allele under insecticide.

Why: Parallel directional change across independently maintained large populations under one exposure, together with a no-insecticide comparison, supports differential survival or reproduction. It does not alone prove the molecular mechanism. A consistent rise in eight large independent populations is less likely under nondirectional drift than under a shared selective pressure. The no-insecticide controls make exposure the plausible difference, while leaving molecular causation to further tests.

Why not the others: (A) Drift is nondirectional across replicates; (B) exposure does not establish directed editing; (D) allele-frequency change does not identify dominance.

Key point: Replication and a control distinguish a selective pattern from chance fluctuations.

Q8 / A Plasmodium stages and host transfer

Answer: (A) P, sporozoites.

Why: Infected female Anopheles mosquitoes inject salivary-gland sporozoites during feeding. These first reach liver cells, later producing stages that infect erythrocytes; gametocytes are taken up by a subsequent mosquito. The arrow from mosquito salivary gland to human liver identifies the inoculated stage. Later erythrocytic replication and gametocyte formation occur inside the human, whereas a feeding mosquito takes up gametocytes from blood.

Why not the others: (B) is an intracellular human stage; (C) follows hepatic multiplication; (D) is the sexual precursor ingested from human blood.

Key point: Follow the arrow across hosts before identifying the stage.

Q9 / B Recombinant colony selection

Answer: (B) A recombinant plasmid disrupted tetR.

Why: Growth on ampicillin indicates functional ampR and plasmid uptake; failure on tetracycline indicates loss of tetR activity in the insert-containing construct. The pair of observations provides both positive and negative selection evidence. Ampicillin selects cells retaining a functional plasmid, so growth there is a positive uptake control. The same colony's tetracycline failure indicates interruption of tetR rather than general cell death.

Why not the others: (A) predicts ampicillin sensitivity; (C) predicts growth on both drugs; (D) predicts loss of ampicillin growth.

Key point: Interpret the two selective plates together.

Q10 / C Trophic transfer in a simple food chain

Answer: (C) 80 kJ.

Given: Starting energy=80,000 kJ, with three transfers at 10% each.

Formula/Principle: Available energy after n transfers is initial energy×(0.10)^n.

Step 1: Herbivores receive 8,000 kJ and secondary consumers 800 kJ.

Step 2: Tertiary consumers receive 80 kJ.

Why: The producer-to-tertiary path contains three, not two, trophic transfers. Energy transfer applies to each arrow: $80{,}000 \rightarrow 8{,}000 \rightarrow 800 \rightarrow 80$ kJ. Respiration and waste at successive levels explain why available energy falls, while the number of arrows fixes the exponent.

Why not the others: (A) stops at herbivores; (B) at secondary consumers; (D) adds a fourth transfer.

Key point: Count arrows between trophic levels.

Q11 / D Independent assortment control

Answer: (D) Approximate independent assortment of the two loci.

Why: A double heterozygote crossed to a double recessive reveals its gamete classes directly. Four nearly equal classes fit the 1:1:1:1 expectation; sampling variation explains small deviations. A double-recessive tester contributes only recessive alleles, revealing the heterozygote's gametes in the offspring phenotypes. Four nearly equal classes are compatible with independent assortment within sampling noise, not proof of zero linkage.

Why not the others: (A) predicts mainly two parental classes; (B) concerns a dihybrid self, not a testcross; (C) has no support from the four equal counts.

Key point: Match expected ratios to the actual cross design.

Q12 / A Latent carrier risk updated by an unaffected first child

Answer: (A) 3/32.

Given: Each unaffected parent has carrier probability $2/3$; segregation in different pregnancies is independent once parental genotypes are fixed.

Formula/Principle: Update the probability that both parents are carriers using the unaffected first child.

Step 1:

$$P(\text{both carriers}) = (2/3)^2 = 4/9.$$

Step 2:

$$P(\text{first unaffected}) = 1 - \frac{4}{9}\frac{1}{4} = \frac{8}{9}.$$

Step 3:

$$P(\text{both carriers} \mid \text{first unaffected}) = \frac{(4/9)(3/4)}{8/9} = \frac{3}{8}.$$

Step 4:

$$P(\text{second affected} \mid \text{first unaffected}) = \frac{3}{8}\frac{1}{4} = \frac{3}{32}.$$

Why: The first child's phenotype provides evidence about the parents' unknown genotypes, even though individual meioses are independent conditional on those genotypes. An unaffected child therefore reduces, but does not eliminate, the chance that both parents carry the recessive allele.

Why not the others: (B) ignores the first child's information; (C) incorrectly assigns carrier probability one half to each parent after conditioning; (D) assumes both parental genotypes are already known to be Aa.

Key point: Separate shared uncertainty about parental genotype from independent segregation.

Q13 / B Cardiac output and stroke volume

Answer: (B) Increase by 0.6 L/min.

Given: Initial HR×SV=60×80 mL/min; final=90×60 mL/min.

Formula/Principle: Cardiac output equals heart rate multiplied by stroke volume.

Step 1: Initial output=4,800 mL/min or 4.8 L/min.

Step 2: Final output=5,400 mL/min or 5.4 L/min; difference=+0.6 L/min.

Why: A fall in stroke volume does not guarantee a fall in total output when rate rises enough. At rest the heart pumps 4.8 L each minute; after the changes it pumps 5.4 L. Compensating factors need not cancel, so multiplying rate and volume in each state is necessary before comparing outputs.

Why not the others: (A) reverses the sign; (C) uses heart-rate change without the volume change; (D) assumes exact compensation.

Key point: Multiply both changing factors before comparing.

Q14 / C Osmosis versus facilitated diffusion

Answer: (C) Net water movement follows a water-potential gradient.

Why: Aquaporins provide a selective path for passive water transport; direction depends on the water-potential difference across the membrane. A channel can increase the rate without reversing the thermodynamic direction. A channel changes membrane permeability but does not supply metabolic energy. The chemical-potential difference of water sets net direction; at equilibrium individual water molecules may still cross both ways without net flow.

Why not the others: (A) reverses the usual osmotic direction; (B) assigns a pump function to a water channel; (D) incorrectly requires ATP for passive flow.

Key point: Separate transport pathway from driving force.

Q15 / D Species-area logarithmic slope

Answer: (D) 1/2.

Given: S(4A)/S(A)=40/20=2.

Formula/Principle: In S=cA^z, the ratio for areas 4A and A is 4^z; c cancels.

Step 1: 4^z=2.

Step 2: Because 4^(1/2)=2, z=1/2.

Why: The exponent measures a log–log slope, not the raw species-count change. The ratio $S(4A)/S(A) = 4^z$ removes unknown c, so the measured doubling gives $4^z = 2$. This exponent is a logarithmic response to area scaling, not a linear increase of one species per area unit.

Why not the others: (A) predicts a fourfold rise; (B) sixteenfold; (C) only √2-fold.

Key point: Divide paired measurements to eliminate the unknown coefficient.

Chemistry

Q16 / A Hybridisation and sigma-pi count

Answer: (A) 6 σ and 3 π.

Why: Three C–H links and the C–C, C–C and C–N connections each supply one σ bond, totalling six. The double bond adds one π and the triple bond adds two π. In a multiple bond the first overlap is sigma and only additional overlaps are pi. Count each C–H connection once and each heavy-atom pair once for sigma, then add one pi from C=C and two from C≡N.

Why not the others: (B) omits one C–H σ; (C) omits a triple-bond π; (D) misallocates one σ in place of π.

Key point: Each bonded atom pair has exactly one σ bond, regardless of bond order.

Q17 / B Oxidation state from complex charge

Answer: (B) +3, 6.

Given: Four neutral ammonia and two −1 chloride ligands give net charge +1.

Formula/Principle: Oxidation state plus ligand charges equals complex charge; coordination number counts donor atoms.

Step 1: x−2=+1, hence x=+3.

Step 2: Six monodentate ligands supply six donor atoms.

Worked derivation:

$$x + 4(0) + 2(-1) = +1 \Rightarrow x = +3, \qquad \text{CN} = 4 + 2 = 6.$$

Why: Formal charge and donor count answer different parts of the question. The charge equation is $x + 4(0) + 2(-1) = +1$, giving $x = +3$. Coordination number counts bound donor atoms, so two chloride donors must be included alongside four ammonia donors.

Why not the others: (A) ignores chloride charge; (C) and (D) count only ammonia donors.

Key point: Count all ligands but include charge only for charged ones.

Q18 / D Formal charge in nitrate resonance

Answer: (D) N +1; each O −1.

Why: N has four bond-order units and no lone pair, assigning formal +1; a singly bonded O has three lone pairs and formal −1. The doubly bonded O is neutral; the total is −1. The octet resonance contributor has nitrogen bonded four times and no lone pair. Formal charge bookkeeping gives +1 on N, −1 on each singly bonded O and zero on doubly bonded O, summing to the ion's −1 charge.

Why not the others: (A) gives net −2; (B) net +1; (C) miscounts N's bonding electrons.

Key point: Check the sum of atom formal charges against ion charge.

Q19 / A Hydrogen emission and orbital occupancy

Answer: (A) 4→2→1.

Given: For emission from initial level n_i to final level n_f, the photon wavenumber is $\widetilde{\nu} = R(n_f^{-2} - n_i^{-2})$.

Formula/Principle: Emission lowers the principal quantum number; the photon energy equals the difference of inverse squares.

Step 1: R(1/4−1/16) corresponds to 4→2.

Step 2: R(1−1/4) corresponds to 2→1, consistent with the intermediate state.

Worked derivation:

$$\widetilde{\nu}_1 = R(1/2^2 - 1/4^2) \Rightarrow 4 \to 2, \quad \widetilde{\nu}_2 = R(1/1^2 - 1/2^2) \Rightarrow 2 \to 1.$$

Why: A proposed sequence must match both energies and the shared intermediate level. Each emitted photon corresponds to an allowed downward energy difference, and the second transition must begin where the first ends. Matching only one line could leave a physically disconnected pair of transitions.

Why not the others: (B) gives 1/4−1/9 first; (C) ends its first transition at 3; (D) gives 1/4−1/25 first.

Key point: Use both spectral lines as a linked path.

Q20 / B Carbocation stability and rearrangement

Answer: (B) The rearrangement can be favourable because hyperconjugation stabilises the tertiary centre.

Why: The migrating H moves with its electron pair from an adjacent carbon; the cation centre shifts to the carbon that lost H. Greater alkyl substitution commonly stabilises a carbocation. The hydride migrates with its bonding electron pair to the electron-deficient centre. The positive charge relocates to the donor carbon, which becomes tertiary in the stated arrangement and is more stabilised by nearby alkyl groups.

Why not the others: (A) reverses the stated stability trend; (C) invokes an external source; (D) violates charge conservation.

Key point: Follow the electron pair and compare the two carbocation environments.

Q21 / C Acyl derivative hydrolysis

Answer: (C) CH_3COOH and HCl.

Why: Water substitutes the acid chloride's Cl at the acyl carbon to give acetic acid; the leaving chloride combines with a proton to give HCl. Carbon skeleton and oxidation state remain those of the acid derivative. Acyl chlorides undergo nucleophilic acyl substitution: water attacks the carbonyl carbon, the tetrahedral intermediate loses chloride and proton transfers yield the acid. No C–C bond needs breaking.

Why not the others: (A) invents an aldehyde and chlorine; (B) breaks the C–C framework; (D) forms a ketone rather than hydrolysing the acyl chloride.

Key point: Hydrolysis replaces the acyl leaving group with OH.

Q22 / A Hinsberg test on amine classes

Answer: (A) Primary amine.

Why: A primary amine yields a sulfonamide retaining N–H; base removes that proton to make a water-soluble salt. Acidification regenerates the less soluble neutral sulfonamide. A primary-amine sulfonamide retains a proton on nitrogen and can form an anionic salt in base. Acidification reprotonates it; a secondary sulfonamide lacks that acidic N–H, so the sequence of solubilities matters.

Why not the others: (B) Secondary-amine sulfonamide lacks N–H and does not show this base solubility; (C) cannot form the same sulfonamide; (D) is permanently charged and not an amine of this class.

Key point: The acid–base behaviour tests whether sulfonamide N–H remains.

Q23 / B Amino acid charge and isoelectric point

Answer: (B) H_3N^+–CHR–COO^-, net 0.

Why: pH 6 lies above the carboxyl pK_a, so carboxyl is largely deprotonated, but below the ammonium pK_a, so nitrogen remains largely protonated. Opposite unit charges cancel. Both relevant pH comparisons must be made independently. At pH 6 the carboxyl group is predominantly COO^- while ammonium is predominantly NH_3^+, producing a zwitterion rather than a neutral uncharged structure.

Why not the others: (A) requires more acidic pH; (C) more basic pH; (D) combines protonation states inconsistent with both pK_a values.

Key point: Compare pH separately with each ionisable group's pK_a.

Q24 / D Arrhenius plot and activation energy

Answer: (D) Y, by a factor of two.

Why: Arrhenius linearisation gives slope $-E_a/R$ for ln k versus 1/T. Y's slope magnitude is twice X's, so its barrier is twice as large when the same R applies. Intercepts affect pre-exponential factors, not the slope comparison. The plot obeys $\ln k = \ln A - E_a/(RT)$, so a slope in kelvin is $-E_a/R$. The larger-magnitude negative slope of Y corresponds to a larger barrier even if its intercept differs.

Why not the others: (A) reverses the slopes; (B) ignores their magnitudes; (C) would apply only without the axis and Arrhenius assumptions.

Key point: Use absolute slope, since the minus sign represents temperature dependence.

Q25 / C Precipitation threshold after dilution by mixing

Answer: (C) No precipitation; $Q = 2.5 \times 10^{-11} < K_{sp}$.

Given: Combining equal volumes doubles the volume available to each initial solute.

Formula/Principle: Compare the post-mixing ionic product $Q = [Ag^+][Cl^-]$ with K_{sp}.

Step 1:

$$[Ag^+]_0 = 5.0 \times 10^{-5}\ \text{M}, \quad [Cl^-]_0 = 5.0 \times 10^{-7}\ \text{M}.$$

Step 2:

$$Q = (5.0 \times 10^{-5})(5.0 \times 10^{-7}) = 2.5 \times 10^{-11}.$$

Step 3: Since $Q < K_{sp}$, the solution is unsaturated and no AgCl precipitates in this model.

Why: The ion concentrations must be evaluated after dilution by mixing, not taken directly from separate stock bottles. A solubility product is an equilibrium threshold; a smaller ionic product does not drive precipitation.

Why not the others: (A) uses undiluted concentrations and mistakes equality for supersaturation; (B) also miscalculates the powers of ten; (D) quotes a product above the threshold but draws the opposite conclusion.

Key point: Account for mixing volumes before testing precipitation.

Q26 / B Disproportionation of chlorine in alkali

Answer: (B) One chlorine atom loses one electron while another gains one electron.

Why: Cl starts at 0 in Cl_2; one product chlorine is −1 in chloride, the other is +1 in hypochlorite. The paired one-electron reduction and oxidation balance internally in disproportionation. One chlorine in hypochlorite is oxidised to +1, while the other in chloride is reduced to −1. The equal one-electron changes can balance without an external oxidant or reductant.

Why not the others: (A) neglects hypochlorite; (C) averages −1 and +1 incorrectly; (D) leaves oxygen at −2.

Key point: Assign oxidation states to each chlorine product separately.

Q27 / D Calorimetric final temperature

Answer: (D) 50 °C.

Given: Both samples have equal mass and specific heat.

Formula/Principle: Heat lost by hot sample equals heat gained by cold sample.

Step 1: mc(80−T)=mc(T−20).

Step 2: 80−T=T−20, giving T=50 °C.

Worked derivation:

$$mc(80-T)=mc(T-20)\Rightarrow 2T=100\Rightarrow T=50^{\circ}\text{C}.$$

Why: Equal thermal capacities make the arithmetic midpoint the energy-balance result. The hot portion loses (mc(80-T)) while the cold gains (mc(T-20)). Because masses and specific heats match, equal heat exchange reduces to equal temperature gaps around the midpoint.

Why not the others: (A) or (B) would require zero change in one sample; (C) gives unequal heat magnitudes.

Key point: The midpoint works only when the heat capacities match.

Q28 / D Raoult partial pressure from mixture

Answer: (D) 2/5.

Given: xA=.25 and xB=.75.

Formula/Principle: Raoult's law gives $p_i=x_ip_i^{\circ}$.

Step 1: pA=.25×80=20 kPa; pB=.75×40=30 kPa.

Step 2: Fraction pA/(pA+pB)=20/50=2/5.

Worked derivation:

$$p_A=0.25(80)=20\ \text{kPa},$$
$$p_B=0.75(40)=30\ \text{kPa},$$
$$y_A=p_A/(p_A+p_B)=20/50=2/5.$$

Why: Vapour composition differs from liquid composition because pure-component pressures differ. Raoult's law weights each pure vapour pressure by its liquid mole fraction. A is only one quarter of liquid molecules but has twice B's pure pressure, so its vapour share is not one quarter.

Why not the others: (A) assumes equal partial pressures; (B) and (C) use an incorrect total or exchange the two vapour shares.

Key point: Calculate both partial pressures before taking a share.

Q29 / A Electrophilic aromatic substitution directing effects

Answer: (A) Replace CH_3 with NO_2.

Why: Nitro withdraws electron density by resonance and induction, destabilising ortho/para sigma complexes relative to meta attack. Methyl donation instead favours ortho/para substitution. Methyl activates and directs ortho/para, whereas nitro strongly withdraws and destabilises the ortho/para carbocation intermediates. Halogen is an important counterexample: it deactivates overall yet still directs ortho/para.

Why not the others: (B) and (C) donate by resonance and direct ortho/para; (D) is deactivating but still ortho/para directing by lone-pair resonance.

Key point: Direction and overall reaction rate are separate substituent effects.

Q30 / C Aryl versus alkyl nucleophilic substitution

Answer: (C) Chlorobenzene's aryl C–Cl bond has partial double-bond character and its carbon is sp^2.

Why: Lone-pair conjugation strengthens the aryl bond, ordinary SN2 approach at sp^2 carbon is disfavoured, and phenyl cation formation is costly. Benzylic carbon is sp^3 and its transition states/intermediates can be resonance stabilised. The aryl carbon is sp^2 and its C–Cl bond is conjugated with the ring, hindering both ordinary backside attack and phenyl-cation formation. Benzylic carbon is sp^3 and can stabilise substitution pathways by resonance.

Why not the others: (A) is false; (B) reverses cation stability; (D) falsely excludes benzylic SN2.

Key point: Compare the carbon bearing chlorine, not simply whether a ring is present.

Mathematics

Q31 / A Hyperbola asymptotes from vertices and focus

Answer: (A) $y=\pm(4/3)x$.

Given: Horizontal transverse axis, $a=3$ and $c=5$.

Formula/Principle: For $x^2/a^2-y^2/b^2=1$, $c^2=a^2+b^2$ and asymptotes are $y=\pm(b/a)x$.

Step 1: $b^2=25-9=16$, so $b=4$.

Step 2: Slopes are $\pm 4/3$.

Worked derivation:

$$a=3,\ c=5,\quad b=\sqrt{c^2-a^2}=\sqrt{25-9}=4;\qquad y=\pm\frac{b}{a}x=\pm\frac{4}{3}x.$$

Why: The focal distance is not itself the conjugate semiaxis. The focus distance c is larger than the transverse semiaxis, and the conjugate semiaxis must be derived before using the asymptote formula.

Why not the others: (B) swaps a and b; (C) uses c in place of b; (D) mixes b with c.

Key point: Recover the missing semiaxis before finding slope.

Q32 / B Common point of a line family

Answer: (B) (7/3,1/3).

Given: The point must satisfy the equation for every real k.

Formula/Principle: Equate separately coefficients of k and constant terms in an identity.

Step 1: k terms give x+2y=3; constants give x−y=2.

Step 2: y=1/3, x=7/3.

Worked derivation:

$$(k+1)x+(2k-1)y-3k-2=k(x+2y-3)+(x-y-2)=0 \quad \forall k.$$

Why: A fixed point must survive any choice of the line parameter, not just one selected k. An identity in k requires both bracketed coefficients to vanish. Solving the resulting two independent lines gives the unique common point.

Why not the others: (A) misses both equations; (C) satisfies x−y=2 but not x+2y=3; (D) misses the constant equation.

Key point: Separate parameter-dependent and independent parts.

Q33 / C Definite integral of rational function

Answer: (C) ln 3.

Given: The denominator factors as (x+1)(x+2).

Formula/Principle: Use partial fractions and evaluate at both bounds.

Step 1: (2x+3)/[(x+1)(x+2)]=1/(x+1)+1/(x+2).

Step 2: [ln(x+1)+ln(x+2)]$_0^1$=ln2+ln3−ln2=ln3.

Worked derivation:

$$\frac{2x+3}{(x+1)(x+2)}=\frac{1}{x+1}+\frac{1}{x+2}, \quad I=[\ln(x+1)+\ln(x+2)]_0^1=\ln 3.$$

Why: Both logarithms contribute boundary terms; the ln2 at the upper bound cancels the ln2 at the lower bound of the other term. The decomposition coefficients are both one because their combined numerator is (x+2)+(x+1). Both lower-bound logarithms must be subtracted.

Why not the others: (A) omits a surviving log; (B) integrates only 1/(x+2); (D) omits the lower endpoint.

Key point: Simplify evaluated logarithms before selecting a result.

Q34 / D Area between parabola and line

Answer: (D) 4/3.

Given: The curves meet at x=0 and x=2.

Formula/Principle: Area is ∫(upper−lower) dx, never a signed cancellation.

Step 1: On (0,2), 2x exceeds x².

Step 2: ∫$_0^2$(2x−x²)dx=[x²−x³/3]$_0^2$=4−8/3=4/3.

Worked derivation:

$$x^2=2x \Longrightarrow x=0,2; \qquad A=\int_0^2 (2x-x^2)\,dx=\left[x^2-\frac{x^3}{3}\right]_0^2=\frac{4}{3}.$$

Why: The linear curve stays above until the second intersection. Checking the sign of 2x−x² between the intersection points determines which graph is above and prevents a signed-area error.

Why not the others: (A) and (B) lose a factor from the interval; (C) drops the subtraction.

Key point: Determine intersections and upper curve first.

Q35 / A Implicit tangent of a level curve

Answer: (A) −4/5.

Given: x^2+xy+y^2 is constant along the curve.

Formula/Principle: Differentiate implicitly, treating y as a function of x.

Step 1: $2x+y+xy'+2yy'=0$, so $y'=-(2x+y)/(x+2y)$.

Step 2: At (1,2), $y'=-4/5$.

Worked derivation:

$$\frac{d}{dx}(x^2+xy+y^2)=2x+y+(x+2y)y'=0, \qquad y'(1,2)=-\frac{2+2}{1+4}=-\frac{4}{5}.$$

Why: The product xy contributes both y and xy′. Differentiating xy needs the product rule because y varies with x along the curve; treating y as a fixed constant would miss xy′.

Why not the others: (B) inverts the derivative; (C) misses the sign; (D) drops the x dependence.

Key point: Apply the product rule to xy.

Q36 / C Equivalence relation modulo a divisor

Answer: (C) 2.

Worked derivation:

$$aRb \iff a \equiv b \pmod 4, \qquad \{\text{odd residues mod } 4\}=\{1,3\}.$$

Why: Congruence modulo 4 is reflexive, symmetric and transitive, so restriction to a subset preserves those properties. Odd integers fall into residues 1 or 3 modulo 4, both nonempty and distinct. Restriction of an equivalence relation to odd integers preserves reflexivity, symmetry and transitivity. Infinite members within each residue do not create extra classes.

Why not the others: (A) merges residues 1 and 3; (B) includes even residue classes absent from the domain; (D) mistakes infinitely many members for infinitely many classes.

Key point: Count equivalence classes on the stated domain, not all integers.

Q37 / D Matrix equation and noncommutativity

Answer: (D) 1.

Given: AX=B, so $X=A^{-1}B$, not BA^{-1}.

Formula/Principle: $A^{-1}=[[1,-1],[0,1]]$.

Step 1: $A^{-1}B=[[0,-1],[1,1]]$.

Step 2: Its trace is 0+1=1.

Worked derivation:

$$A^{-1}=\begin{pmatrix}1 & -1\\ 0 & 1\end{pmatrix}, \quad X=A^{-1}B=\begin{pmatrix}0 & -1\\ 1 & 1\end{pmatrix}, \qquad \operatorname{tr} X=0+1=1.$$

Why: Left multiplication preserves the order dictated by AX=B; matrix multiplication is not commutative. Premultiplication by A^{-1} cancels A in AX=B. Multiplying in the opposite order can change the matrix even if a trace occasionally coincides.

Why not the others: (A) adds off-diagonal entries; (B) uses the diagonal of B; (C) omits the lower-right entry.

Key point: Solve the matrix equation on the correct side.

Q38 / B Conditional coin-run counting

Answer: (B) 3/5.

Given: Conditional outcomes are the $\binom{6}{3} = 20$ equally likely placements of three heads.

Formula/Principle: Exactly one HH pair means one head-run of length two and one head-run of length one.

Step 1: Place the three tails first. They create four gaps: before, between and after the tails.

Step 2: Choose two distinct gaps for the head-runs and choose which gets HH:

$$N = \binom{4}{2} 2 = 12.$$

Step 3:

$$P = \frac{12}{20} = \frac{3}{5}.$$

Worked derivation:

$$P(\text{one HH pair} \mid \text{three heads}) = \frac{2\binom{4}{2}}{\binom{6}{3}} = \frac{3}{5}.$$

Why: Putting the runs in different tail-separated gaps prevents a second HH pair. Conditioning on three heads changes the denominator from all 64 toss sequences to the 20 allowed sequences.

Why not the others: (A) undercounts run placements; (C) counts the complement, containing zero or two HH pairs; (D) includes outcomes with additional head adjacency.

Key point: Translate adjacency into runs before counting conditional outcomes.

Q39 / C Compound-angle cosine under quadrant constraints

Answer: (C) 1/√2.

Given: Acute angles give positive sine and cosine.

Formula/Principle: cos(α+β)=cosα cosβ−sinα sinβ.

Step 1: (cosα,sinα)=(2,1)/√5 and (cosβ,sinβ)=(3,1)/√10.

Step 2: (6−1)/√50=5/√50=1/√2.

Worked derivation:

$$\cos\alpha = \frac{2}{\sqrt{5}},\ \sin\alpha = \frac{1}{\sqrt{5}},\ \cos\beta = \frac{3}{\sqrt{10}},\ \sin\beta = \frac{1}{\sqrt{10}};\quad \cos(\alpha+\beta) = \frac{6-1}{\sqrt{50}} = \frac{1}{\sqrt{2}}.$$

Why: Quadrant information fixes the signs, while the subtraction reflects the compound angle. Acute-angle information fixes all component signs positive. The compound-angle identity still subtracts the sine product even though each sine is positive.

Why not the others: (A) and (B) omit a term; (D) ignores β.

Key point: Build unit-triangle components before combining angles.

Q40 / D Foot on coordinate plane

Answer: (D) (2,−3,0).

Given: The xy-plane consists of z=0.

Formula/Principle: An orthogonal projection onto that plane changes only the normal (z) coordinate.

Step 1: Keep x=2 and y=−3.

Step 2: Set z=0.

Worked derivation:

$$\Pi_{z=0}(2,-3,5) = (2,-3,0), \qquad (2,-3,5) - (2,-3,0) = (0,0,5) \perp \{z=0\}.$$

Why: The connecting segment is parallel to the z-axis and perpendicular to the plane. The connecting vector is normal to every direction within the xy-plane, which verifies that the projected point is the perpendicular foot.

Why not the others: (A) and (B) change a coordinate within the plane; (C) changes both in-plane coordinates.

Key point: Identify the plane's normal direction.

Q41 / A Cross product from area and orientation

Answer: (A) (0,0,−3).

Given: Both vectors lie in the xy-plane.

Formula/Principle: The z-component of a×b is $a_x b_y - a_y b_x$.

Step 1: 1×1−2×2=−3.

Step 2: The other components vanish, giving $-3\hat{\mathbf{k}}$.

Worked derivation:

$$\mathbf{a}\times\mathbf{b}=\begin{vmatrix}\hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}}\\ 1 & 2 & 0\\ 2 & 1 & 0\end{vmatrix}=(1-4)\hat{\mathbf{k}}=-3\hat{\mathbf{k}}.$$

Why: Magnitude 3 is area, but order a×b sets the negative orientation. The determinant encodes oriented area, so changing the vector order reverses the sign while keeping area magnitude three.

Why not the others: (B) reverses order; (C) is in-plane; (D) uses the dot-product sum.

Key point: Preserve operand order in cross products.

Q42 / C Cube roots of unity as geometric points

Answer: (C) 3.

Given: ω=−1/2±i√3/2, a vertex of the unit-circle equilateral triangle.

Formula/Principle: $|z|^2 = z\,\bar{z}$, or sum of squared real and imaginary parts.

Step 1: 1−ω=3/2∓i√3/2.

Step 2: Modulus squared=9/4+3/4=3.

Worked derivation:

$$\omega=-\tfrac{1}{2}\pm i\tfrac{\sqrt{3}}{2},\quad |1-\omega|^2=\left(\tfrac{3}{2}\right)^2+\left(\tfrac{\sqrt{3}}{2}\right)^2=3.$$

Why: Both non-real conjugate choices have the same distance from 1. The two non-real cube roots are complex conjugates; both give the same squared distance from the real point 1.

Why not the others: (A) uses $|\omega|^2$; (B) omits the real shift; (D) would correspond to opposite unit-circle points.

Key point: Geometric distance is invariant under conjugation here.

Q43 / D Two positive real roots

Answer: (D) t>4.

Given: Product of roots=4>0 and sum=t.

Formula/Principle: Distinct real roots require discriminant $t^2-16>0$; positive sum selects positive rather than negative roots.

Step 1: Real distinct gives t<−4 or t>4.

Step 2: Positive sum removes t<−4, leaving t>4.

Worked derivation:

$$r_1+r_2=t>0,\quad r_1r_2=4>0,\quad \Delta=t^2-16>0\Longrightarrow t>4.$$

Why: At t=4 the two positive roots coincide, violating distinctness. Positive product alone allows

both roots negative, so the sum fixes their sign. The strict discriminant excludes the repeated positive root at t=4.

Why not the others: (A) admits non-real roots; (B) includes the double root; (C) gives two negative roots.

Key point: Combine discriminant and Vieta signs with strict endpoints.

Q44 / A Optimising a constant-term coefficient across exponents

Answer: (A) 2.

Given: Choosing r factors of x^(−p) gives power 2(6−r)−pr=12−(p+2)r.

Formula/Principle: A constant term needs integer r=12/(p+2) between 0 and 6; its coefficient is C(6,r).

Step 1: For p=1,2,4,10, the allowed r values are 4,3,2,1.

Step 2: Corresponding coefficients are 15,20,15,6, so p=2 gives the maximum.

Worked derivation:

$$T_r = \binom{6}{r} x^{12-(p+2)r}, \quad r = \frac{12}{p+2}; \quad p = 1, 2, 4, 10 \Rightarrow r = 4, 3, 2, 1 \Rightarrow \binom{6}{r} = 15, 20, 15, 6.$$

Why: Exponent cancellation alone does not determine which of several valid expansions has the largest coefficient. Each listed p gives a different valid constant term. Comparing coefficients after solving the exponent condition shows why the central binomial choice wins.

Why not the others: (B) and (C) each give 15; (D) gives 6.

Key point: Compare the binomial coefficients only after finding each admissible constant term.

Q45 / B Geometric-series convergence boundary

Answer: (B) |r|<1.

Worked derivation:

$$S_n = \frac{1 - r^{n+1}}{1 - r} \quad (r \neq 1), \qquad \lim_{n\to\infty} r^{n+1} = 0 \Longleftrightarrow |r| < 1.$$

Why: The nth partial sum is (1−rⁿ)/(1−r) for r≠1. It approaches 1/(1−r) only when rⁿ→0, precisely for −1<r<1; r=1 grows and r=−1 oscillates. The finite-sum identity alone does not establish infinite convergence. At r=−1 partial sums oscillate and at r=1 they grow without bound.

Why not the others: (A) and (D) each admit ratios of magnitude at least one; (C) incorrectly includes both boundary values.

Key point: Test endpoints rather than silently extending a finite-series formula.

Physics

Q46 / A Maximum fractional uncertainty in resistance

Answer: (A) 3%.

Given: Fractional maximum bounds are 0.02 in V and 0.01 in I.

Formula/Principle: First-order worst-case relative errors add for a quotient.

Step 1: δR/R≈δV/V−δI/I as signed perturbations.

Step 2: Maximum magnitude≤2%+1%=3%.

Worked derivation:

$$R = \frac{V}{I}, \quad \frac{dR}{R} \approx \frac{dV}{V} - \frac{dI}{I}, \quad \left|\frac{\delta R}{R}\right|_{\max} \leq 0.02 + 0.01 = 0.03.$$

Why: Opposite signed errors can reinforce in the quotient; subtracting percentages is not a safe upper bound. For a worst-case bound, voltage and current errors can have opposite signs and reinforce in the quotient. This is not the root-sum-square rule for independent random standard deviations.

Why not the others: (B) ignores current; (C) ignores voltage; (D) multiplies percentage numbers.

Key point: Distinguish worst-case uncertainty from a specific signed deviation.

Q47 / B Static friction at threshold

Answer: (B) 20 N.

Given: Normal reaction=mg=50 N on the horizontal floor.

Formula/Principle: Limiting static friction f_s,max=μsN.

Step 1: f_s,max=.4×50=20 N.

Step 2: At the threshold the applied force balances this maximum.

Worked derivation:

$$N = mg = (5)(10) = 50\,\text{N}, \qquad f_{s,\text{max}} = \mu_s N = (0.4)(50) = 20\,\text{N}.$$

Why: Static friction adjusts to smaller forces before its limit is reached. Below 20 N static friction merely matches the applied force; the formula gives its maximum available magnitude at the threshold, not its value under every smaller pull.

Why not the others: (A) halves the result; (C) uses N itself; (D) omits the mass factor.

Key point: Use the limiting value only at impending motion.

Q48 / D Series extension of equal wires

Answer: (D) 3FL/(2AY).

Given: Both series wires carry the same tension F.

Formula/Principle: Each extension equals FL/(AY_i); series extensions add.

Step 1: First wire extends FL/(AY), second FL/(2AY).

Step 2: Sum=3FL/(2AY).

Worked derivation:

$$\Delta L_1 = \frac{FL}{AY}, \quad \Delta L_2 = \frac{FL}{A(2Y)}, \quad \Delta L_{\text{total}} = \Delta L_1 + \Delta L_2 = \frac{3FL}{2AY}.$$

Why: Equal force does not imply equal strain for different moduli. The junction transmits the same force through both series elements, while their displacements add; using an average modulus directly would obscure this force balance.

Why not the others: (A) retains only the stiffer wire; (B) only the first; (C) assumes identical moduli.

Key point: Add displacements for elements in series.

Q49 / C Differential thermal expansion in a ring-gap arrangement

Answer: (C) The sphere can no longer pass through the hole.

Worked derivation:

$$d_{\text{hole}} = d(1 + \alpha\Delta T), \qquad d_{\text{sphere}} = d(1 + 2\alpha\Delta T), \quad d_{\text{sphere}} - d_{\text{hole}} = d\alpha\Delta T > 0.$$

Why: A hole expands as if it were made of the plate material; its diameter scales by approximately 1+αΔT, while the sphere diameter scales by 1+2αΔT. Starting equal, the sphere becomes larger. Heating a plate expands its hole just as an imaginary plug of the plate material would expand. Because the sphere has twice the coefficient, its diameter outgrows the hole.

Why not the others: (A) falsely makes a heated hole contract; (B) ignores the specified coefficients; (D) reverses their ratio.

Key point: Apply each material's linear expansion to its own diameter.

Q50 / D Equipartition and internal energy

Answer: (D) $2T_0$.

Given: Diatomic gas has five active quadratic degrees of freedom, n=1, and work on gas=$5RT_0/2$.

Formula/Principle: U=(5/2)nRT and first law ΔU=Q+W_on.

Step 1: Q=0 and $\Delta U=(5/2)R(T_f-T_0)=5RT_0/2$.

Step 2: $T_f-T_0=T_0$, hence $T_f=2T_0$.

Worked derivation:

$$\Delta U=\frac{5}{2}R(T_f-T_0)=Q+W_{\text{on}}=0+\frac{5}{2}RT_0\Longrightarrow T_f=2T_0.$$

Why: Mechanical stirring raises internal energy despite the container being thermally insulated. The rigid container prevents boundary-volume work but not paddle work. The five active degrees of freedom set internal-energy change under the stated vibrational approximation.

Why not the others: (A) halves the temperature rise; (B) mistakes work in units RT for final T; (C) doubles the rise.

Key point: Insulated means Q=0, not ΔU=0.

Q51 / B Pendulum period under length change

Answer: (B) 1.20.

Given: Lnew/Lold=1.44.

Formula/Principle: $T=2\pi\sqrt{(L/g)}$ for a simple pendulum at small amplitude.

Step 1: Tnew/Told=√1.44.

Step 2: Ratio=1.20.

Worked derivation:

$$\frac{T_{\text{new}}}{T_{\text{old}}}=\sqrt{\frac{L_{\text{new}}}{L_{\text{old}}}}=\sqrt{1.44}=1.20.$$

Why: Period varies with square root of length rather than length itself. Both observations use the same g and small-amplitude regime, so those common factors cancel. A 44% length rise causes only a 20% period rise.

Why not the others: (A) uses direct proportionality; (C) takes reciprocal; (D) squares length ratio.

Key point: Cancel common factors before taking the square root.

Q52 / A Beat-frequency ambiguity with two references

Answer: (A) 252 Hz.

Given: |f−256|=4 and |f−260|=8.

Formula/Principle: Beat frequency is the absolute difference of two close frequencies.

Step 1: First reference permits f=252 or 260 Hz.

Step 2: Against 260 Hz, these give 8 or 0 beats/s; only 252 qualifies.

Worked derivation:

$$|f-256|=4\Rightarrow f\in\{252,260\};\qquad |f-260|=8\Rightarrow f\in\{252,268\};\quad \text{intersection}=\{252\}\,\text{Hz}.$$

Why: One beat count alone has a high/low ambiguity. Each beat measurement produces two possible frequencies. Only intersecting both candidate sets removes the high/low ambiguity.

Why not the others: (B) gives 8 beats against 256; (C) gives none against 260; (D) gives none against 256.

Key point: Intersect candidate sets from two reference tones.

Q53 / D Potential difference across a uniform field

Answer: (D) 4 V.

Given: E_x=+3 V/m over 2 m.

Formula/Principle: $\Delta V = -\int E_x\,dx$.

Step 1: V(2)−V(0)=−3×2=−6 V.

Step 2: V(2)=10−6=4 V.

Worked derivation:

$$V(2) - V(0) = -\int_0^2 E_x\,dx = -(3\,\mathrm{V/m})(2\,\mathrm{m}) = -6\,\mathrm{V}, \quad V(2) = 4\,\mathrm{V}.$$

Why: Electric potential decreases along the field direction for positive E. Potential falls in the direction of a positive field. The integration limits match the region where the field is specified.

Why not the others: (A) reverses sign; (B) ignores field; (C) uses only one metre.

Key point: Keep the minus sign between field and potential gradient.

Q54 / B Drift speed and current density

Answer: (B) J.

Given: n′=n/2 and v′=2v.

Formula/Principle: Drift current-density magnitude J=nqv.

Step 1: J′=(n/2)q(2v).

Step 2: Factors cancel, so J′=J.

Worked derivation:

$$J' = n'qv' = \left(\frac{n}{2}\right) q(2v) = nqv = J.$$

Why: Fewer carriers can yield the same flow per area if each drifts proportionally faster. This compares magnitudes with the same carrier charge; a faster drift exactly offsets half as many carriers per unit volume.

Why not the others: (A) divides twice; (C) changes only v; (D) compounds two increases that are not both present.

Key point: Current density depends on the product of density and drift speed.

Q55 / C Energy exchange between a capacitor and inductor

Answer: (C) 60 V.

Given: There is no resistor, external supply or mechanical parameter change, so total stored electric-plus-magnetic energy is conserved.

Formula/Principle:

$$U_C = \tfrac{1}{2}CV^2, \quad U_L = \tfrac{1}{2}LI^2, \quad U_C + U_L = U_0.$$

Step 1:

$$U_0 = \tfrac{1}{2}(50 \times 10^{-6})(100)^2 = 0.250\ \mathrm{J}.$$

Step 2:

$$U_L = \tfrac{1}{2}(2.0)(0.40)^2 = 0.160\ \mathrm{J}.$$

Step 3: Remaining capacitor energy is $U_C = 0.250 - 0.160 = 0.090$ J.

Step 4:

$$|V| = \sqrt{\frac{2U_C}{C}} = \sqrt{\frac{0.180}{50 \times 10^{-6}}} = \sqrt{3600} = 60 \text{ V}.$$

Worked derivation:

$$\tfrac{1}{2}C(100)^2 = \tfrac{1}{2}C|V|^2 + \tfrac{1}{2}L(0.40)^2 \Rightarrow |V| = 60 \text{ V}.$$

Why: Energy can move between the electric and magnetic stores while their sum stays fixed in the stated ideal circuit. The energy equation fixes voltage magnitude but not its sign at an unspecified instant, which is why the question asks for a magnitude.

Why not the others: (A) subtracts amplitudes instead of energies; (B) uses an incorrect energy fraction; (D) leaves capacitor energy unchanged despite nonzero inductor energy.

Key point: Balance both field-energy stores before solving for an instantaneous amplitude.

Q56 / D Resonance curve and bandwidth

Answer: (D) P has smaller resistance and smaller bandwidth.

Worked derivation:

$$I_{\text{max}} = \frac{V}{R}, \qquad \Delta\omega = \frac{R}{L}; \quad R_P < R_Q \Longrightarrow I_{\text{max},P} > I_{\text{max},Q},\ \Delta\omega_P < \Delta\omega_Q.$$

Why: At series resonance impedance is R, so peak current V/R is larger for smaller R. Bandwidth in angular frequency is R/L, so the same smaller R narrows the peak with L fixed. The common L and C keep resonance at the same ideal centre frequency. Reading height and width together identifies resistance without mistaking the graphs for different capacitors.

Why not the others: (A) and (B) predict a shorter peak for greater R; (C) conflicts with the common L and C setting the same ideal resonance.

Key point: Read both peak height and width, not one feature alone.

Q57 / A Prism minimum deviation and refractive index

Answer: (A) √2.

Given: Prism angle A=60°, minimum deviation δm=30°.

Formula/Principle: n=sin[(A+δm)/2]/sin(A/2).

Step 1: Numerator sin45°=1/√2 and denominator sin30°=1/2.

Step 2: Ratio=√2.

Worked derivation:

$$n = \frac{\sin[(A + \delta_{\text{min}})/2]}{\sin(A/2)} = \frac{\sin 45^\circ}{\sin 30^\circ} = \frac{\sqrt{2}/2}{1/2} = \sqrt{2}.$$

Why: At minimum deviation the two refractions are symmetric, giving the stated formula. At minimum deviation the ray path is symmetric through the prism, justifying the special relation; substituting the full angles rather than half-angles would be wrong.

Why not the others: (B) uses an incorrect refraction angle; (C) neglects sin45°; (D) inverts the ratio.

Key point: Half both the sum and the prism angle.

Q58 / B Thin-film reflected phase reversals

Answer: (B) 2nt=mλ₀ because both reflections reverse phase.

Worked derivation:

$$\Delta\phi_{\text{reflection}} = \pi - \pi = 0, \qquad \Delta_{\text{optical}} = 2nt, \quad \Delta_{\text{optical}} = m\lambda_0 \text{ for constructive reflection.}$$

Why: Reflection from air to film and from film to higher-index glass each adds a π phase reversal, so their relative reflection phase is zero. The remaining round-trip optical path is 2nt and must equal an integer free-space wavelength. Both interface reflections are from lower to higher refractive index and each reverses phase. Only their relative phase matters, leaving the round-trip path condition.

Why not the others: (A) assumes only one reversal; (C) omits the round trip and index; (D) ignores path-dependent phase.

Key point: Count relative, not absolute, reflection phase changes.

Q59 / C Electron de Broglie wavelength under acceleration

Answer: (C) λ/2.

Given: Electron charge and mass are unchanged; kinetic energy eV.

Formula/Principle: λ=h/√(2meV) in the nonrelativistic limit.

Step 1: Voltage grows by factor four, so momentum grows by √4=2.

Step 2: Wavelength halves.

Worked derivation:

$$\tfrac{1}{2}mv^2 = eV, \quad p = \sqrt{2meV}, \quad \lambda = \frac{h}{p} \propto V^{-1/2}; \qquad \frac{\lambda(4V)}{\lambda(V)} = \frac{1}{2}.$$

Why: Accelerating potential sets kinetic energy, not momentum directly. The nonrelativistic kinetic-energy assumption is needed for p=√(2meV). Higher voltage raises momentum by √4, not by four.

Why not the others: (A) and (B) increase λ with voltage; (D) scales λ inversely with V rather than √V.

Key point: Keep the square-root dependence.

Q60 / C Half-wave rectified mean power

Answer: (C) V₀²/(4R).

Given: Power=v²/R for half a cycle and zero for half a cycle.

Formula/Principle: Average sin² over a positive half-cycle is 1/2.

Step 1: Mean over that active half-cycle is V₀²/(2R).

Step 2: Only half the full period is active, yielding V₀²/(4R).

Worked derivation:

$$\bar{P} = \frac{1}{2\pi}\int_0^{\pi} \frac{V_0^2 \sin^2\theta}{R}\, d\theta = \frac{V_0^2}{2\pi R}\cdot\frac{\pi}{2} = \frac{V_0^2}{4R}.$$

Why: Mean power uses squared voltage, not the square of average voltage. The average is taken over the full period, including the blocked half-cycle. Squaring the average rectified voltage would be a different quantity.

Why not the others: (A) uses peak power; (B) averages only the conducting half; (D) introduces an extra factor two.

Key point: Average over the full period.
